

### PREFACE FOR FACULTY

Linear Time-Invariant (LTI) systems form the core analytical foundation of DSP. This lecture covers system classification, the convolution sum derivation from the sifting property, causality and BIBO stability criteria, and the formulation of Linear Constant-Coefficient Difference Equations (LCCDE).

#### Pedagogical Strategy:

1. Derive the convolution sum starting from the impulse decomposition  $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$ .
2. Demonstrate the 4 operational steps of discrete convolution: **Flip, Shift, Multiply, Sum**.
3. Prove the Bounded-Input Bounded-Output (BIBO) stability condition:  $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$ .
4. Establish the causality condition for an LTI system:  $h[n] = 0$  for all  $n < 0$ .
5. Relate LCCDEs to total system response (homogeneous natural response + particular forced response).

### 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Test** discrete systems for Linearity, Time-Invariance, Causality, Stability, and Memory.
2. **Derive** and **compute** the convolution sum  $y[n] = x[n] * h[n]$  analytically, tabularly, and graphically.
3. **Determine** the duration and sample support of the convolution of two finite-length sequences.
4. **Evaluate** BIBO stability and causality directly from a system's impulse response  $h[n]$ .
5. **Formulate** and **solve** difference equations describing discrete-time recursive and non-recursive systems.

### 2. MATHEMATICAL FOUNDATIONS

#### 2.1 The Convolution Sum Derivation

Any discrete-time input  $x[n]$  can be represented as a weighted sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Let  $\mathcal{T} \cdot$  represent a Linear Time-Invariant operator. Applying  $\mathcal{T}$  to  $x[n]$ :

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By Linearity (Superposition principle):

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\mathcal{T}\delta[n-k]$$

Let  $h[n] = \mathcal{T}\delta[n]$  be the **impulse response**. By Time-Invariance:

$$\mathcal{T}\delta[n-k] = h[n-k]$$

Thus, we arrive at the **Discrete Convolution Sum**:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

## 2.2 Properties of Convolution

1. **Commutativity:**  $x[n] * h[n] = h[n] * x[n]$ .
2. **Associativity:**  $(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$ .
3. **Distributivity:**  $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$ .
4. **Identity:**  $x[n] * \delta[n] = x[n]$ .
5. **Shift Property:**  $x[n - n_1] * h[n - n_2] = y[n - n_1 - n_2]$ .

## 2.3 Length of Convolved Sequences

If sequence  $x[n]$  has length  $L_x$  (from  $n = N_1$  to  $N_2$ ) and impulse response  $h[n]$  has length  $L_h$  (from  $n = M_1$  to  $M_2$ ), the convolved output  $y[n] = x[n] * h[n]$  has:

- Total length:  $L_y = L_x + L_h - 1$ .
- Starting index:  $n_{\text{start}} = N_1 + M_1$ .
- Ending index:  $n_{\text{end}} = N_2 + M_2$ .

## 2.4 Causality and BIBO Stability Criteria

- **Causality:** An LTI system is causal if and only if its impulse response is zero for all negative time:

$$h[n] = 0, \quad \forall n < 0$$

- **BIBO Stability:** An LTI system is Bounded-Input Bounded-Output (BIBO) stable if and only if its impulse response is **absolutely summable**:

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

*Proof:* Let input  $|x[n]| \leq M_x < \infty$  for all  $n$ .

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \leq M_x \sum_{k=-\infty}^{\infty} |h[k]| = M_x \cdot S$$

If  $S < \infty$ , then  $|y[n]| \leq M_y = M_x S < \infty$ .

## 2.5 Linear Constant-Coefficient Difference Equations (LCCDE)

An  $N^{\text{th}}$ -order LCCDE relating input  $x[n]$  to output  $y[n]$  is given by:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m], \quad a_0 = 1$$
$$y[n] = \sum_{m=0}^M b_m x[n-m] - \sum_{k=1}^N a_k y[n-k]$$

The total solution consists of the Homogeneous solution  $y_h[n]$  (natural response satisfying  $\sum a_k y[n-k] = 0$ ) and the Particular solution  $y_p[n]$  (forced response).

## 3. WORKED NUMERICAL EXAMPLES

### Example 2.1: Analytical Convolution of Two Sequences

**Problem:** Compute the linear convolution  $y[n] = x[n] * h[n]$  for:

$$x[n] = a^n u[n], \quad h[n] = b^n u[n], \quad a \neq b, \quad |a| < 1, \quad |b| < 1$$

**Solution:**

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} (a^k u[k])(b^{n-k} u[n-k])$$

The step functions require:

- $u[k] = 1 \implies k \geq 0$ .
- $u[n - k] = 1 \implies k \leq n$ . Thus, for  $n < 0$ , the limits cannot be satisfied and  $y[n] = 0$ . For  $n \geq 0$ , the summation limits are  $k = 0$  to  $n$ :

$$y[n] = \sum_{k=0}^n a^k b^{n-k} = b^n \sum_{k=0}^n \left(\frac{a}{b}\right)^k$$

Using the finite geometric series sum  $\sum_{k=0}^n r^k = \frac{1-r^{n+1}}{1-r}$ :

$$y[n] = b^n \frac{1 - (a/b)^{n+1}}{1 - (a/b)} = b^n \frac{1 - a^{n+1}b^{-(n+1)}}{(b-a)/b} = \frac{b^{n+1} - a^{n+1}}{b-a} u[n]$$

$$y[n] = \left( \frac{b^{n+1} - a^{n+1}}{b-a} \right) u[n]$$

## Example 2.2: Stability and Causality Testing

**Problem:** Determine if the systems with the following impulse responses are causal and BIBO stable: (a)

$$h_1[n] = (0.5)^n u[n+2] \text{ (b) } h_2[n] = 3^n u[-n-1] \text{ (c) } h_3[n] = \left(\frac{1}{n}\right) u[n-1]$$

**Solution:** (a) Causality:  $h_1[-2] = (0.5)^{-2} = 4 \neq 0 \implies$  **Non-causal**. Stability:

$$\sum_{n=-\infty}^{\infty} |h_1[n]| = \sum_{n=-2}^{\infty} (0.5)^n = 4 + 2 + \sum_{n=0}^{\infty} (0.5)^n = 6 + \frac{1}{1-0.5} = 6 + 2 = 8 < \infty \implies \text{BIBO Stable.}$$

(b) Causality:  $h_2[n]$  is non-zero for  $n \leq -1 \implies$  **Non-causal** (anti-causal). Stability: Let  $m = -n$ :

$$\sum_{n=-\infty}^{\infty} |h_2[n]| = \sum_{n=-\infty}^{-1} 3^n = \sum_{m=1}^{\infty} 3^{-m} = \sum_{m=1}^{\infty} \left(\frac{1}{3}\right)^m = \frac{1/3}{1-1/3} = \frac{1/3}{2/3} = 0.5 < \infty \implies \text{BIBO Stable.}$$

(c) Causality:  $h_3[n] = 0$  for  $n < 1 \implies$  **Causal**. Stability:

$$\sum_{n=-\infty}^{\infty} |h_3[n]| = \sum_{n=1}^{\infty} \frac{1}{n}$$

This is the harmonic series, which diverges ( $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ ). Therefore, the system is **Unstable**.

## 4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

### Question 1 (15 Marks)

(a) State and prove the necessary and sufficient condition for a discrete-time LTI system to be BIBO stable. (6 Marks) (b)

Convolve the two finite sequences using the tabular method:

$$x[n] = \underset{\uparrow}{2}, 1, -1, 3, \quad h[n] = \underset{\uparrow}{1}, 2, 0, -2$$

Verify the result using polynomial multiplication. (9 Marks)

**Model Answer & Step-by-Step Marking Rubric:**

- **Part (a):**
  - Statement of condition:  $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$  (2 Marks)
  - Proof of sufficiency: Let  $|x[n]| \leq M_x$ , prove  $|y[n]| \leq M_x \sum |h[k]| < \infty$  (2 Marks)
  - Proof of necessity: Construct bounded input  $x[n] = \text{sgn}(h[-n])$  showing  $y[0] = \sum |h[k]| = \infty$  if divergent (2 Marks)
- **Part (b):**
  - Tabular / Matrix setup:

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(3 Marks)

- Diagonal summation:

- $y[0] = 2$
- $y[1] = 1 + 4 = 5$
- $y[2] = -1 + 2 + 0 = 1$
- $y[3] = 3 - 2 + 0 - 4 = -3$
- $y[4] = 6 + 0 - 2 = 4$
- $y[5] = 0 + 2 = 2$
- $y[6] = -6$

$$y[n] = \underset{\uparrow}{2}, 5, 1, -3, 4, 2, -6, \quad \text{Length } L_y = 4 + 4 - 1 = 7 \text{ samples}$$

(4 Marks)

- Verification by polynomial multiplication:

$$(2 + z^{-1} - z^{-2} + 3z^{-3})(1 + 2z^{-1} - 2z^{-3}) = 2 + 5z^{-1} + z^{-2} - 3z^{-3} + 4z^{-4} + 2z^{-5} - 6z^{-6} \quad (2 \text{ Marks})$$

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## 5. PYTHON VERIFICATION SCRIPT

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import numpy as np

x = np.array([2, 1, -1, 3])
h = np.array([1, 2, 0, -2])
y = np.convolve(x, h)

print(f"Convolution Output y[n] = {y}")
# Expected: [ 2,  5,  1, -3,  4,  2, -6]
```